

INTERSECTIONS OF CAYLEY PERMUTATIONS, FUBINI RANKINGS, AND UNIT INTERVAL PARKING FUNCTIONS.

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ABSTRACT. We characterize and enumerate the intersections of three subsets of parking functions: Cayley permutations, Fubini rankings, and unit interval parking functions. We show that their pairwise intersections share cardinality and can be characterized by the behavior of their weakly increasing elements.

1. INTRODUCTION.

In 1966 Konheim and Weiss introduced parking functions to model hashing tables for computer science. [Melissa: what are “linear probing resolution strategies for random hashing functions”? See Permutohedron paper.](#)

Definition 1.1. *Parking Function* A parking function, $\alpha = (a_1, a_2, \dots, a_n)$.

We discuss three subsets of parking functions which share the following characteristics. [Melissa: Bijections from last summer can be part of the motivation.](#)

Definition 1.2. *Cayley Permutations*

A Cayley Permutation of length n is an n -tuple $\gamma = (c_1, c_2, \dots, c_n)$ such that if $i \in [n]$ appear as entries in γ , then all elements $1 \leq j \leq i$ also appear in γ . We denote the set of all Cayley permutations of length n by $\mathfrak{C}_n$.

Definition 1.3. *Fubini Ranking*

The n -tuple $\beta = (b_1, b_2, \dots, b_n) \in [n]^n$ is a Fubini ranking of length n if the following holds: for all $x \in [n]$, if $k > 0$ entries of β are equal to x , then the next largest value in β is $x + k$. denote the set of Fubini rankings of length n by FR_n .

Definition 1.4. *Unit Interval Parking Functions*

Consider the n -tuple $\alpha = (a_1, a_2, \dots, a_n) \in [n]^n$. Using the standard parking process, car i drives to its preference a_i . and if that parking spot is available, it parks there. Otherwise, it attempts to park in spot $a_i + 1$. If that spot is available, then parking is successful. Otherwise, car i fails to park. If all cars are able to park using this rule, then we say α is a unit interval parking function of length n . A full characterization can be found in [2, Theorem 2.9].

Definition 1.5. *Unit Interval Parking Functions, The Block Definitions*

[2, Definition 2.7] Given $\alpha = (a_1, \dots, a_n) \in \text{PF}_n$, let $\alpha^\uparrow = (a'_1, \dots, a'_n)$ be its weakly increasing rearrangement. The partition of $\alpha^\uparrow$ into a concatenation $\alpha' = \pi_1 | \pi_2 | \dots | \pi_m$, where each block π_j begins at (and includes) the j -th entry satisfying $a'_i = i$, is called the block structure of α . Each π_j is called a block of α . A new block starts at every position i where the sorted value satisfies $a'_i = i$.

For example, if $\alpha = (8, 1, 5, 5, 1, 2, 4, 7) \in PF_8$, then $\alpha^\uparrow = (1, 1, 2, 4, 5, 5, 7, 8)$ and its block structure is $\alpha' = 112|4|55|7|8$.

Lemma 1.1. *UPF_n Canonical Block Form*

Let $\alpha \in UPF_n$ with block structure $\pi_1|\pi_2|\cdots|\pi_m$ as in Definition 1.5. For each $j \in [m]$, let $s = \min(\pi_j)$ and $\ell_j = |\pi_j|$. Then

$$\pi_j = (s, s, s+1, s+2, \dots, s+\ell_j-2).$$

Proof. Fix a block π_j . Let $s = \min(\pi_j)$ and $\ell_j = |\pi_j|$. Index the entries within π_j by $i^* = 1, 2, \dots, \ell_j$. Since the block begins at position s in $\alpha^\uparrow$, the entry in position i^* within the block is a'_{s+i^*-1} . We prove by induction that

i^*	1	2	3	$\dots$	ℓ_j
a'_{s+i^*-1}	s	s	$s+1$	$\dots$	$s+\ell_j-2$

so the first two entries are both s , and each later entry increases by exactly 1.

Recall that by Definition 1.5, a new block begins at position i precisely when $a'_i = i$. Hence, $a'_s = s$. If $\ell_j = 1$, then $\pi_j = (s)$, so assume $\ell_j \geq 2$. For $2 \leq r \leq \ell_j$, let $P(r)$ denote the statement that the first r entries follow the pattern displayed above.

We first prove $P(2)$, namely $(a'_s, a'_{s+1}) = (s, s)$. Since $\alpha^\uparrow$ is weakly increasing and α is a parking function, $s = a'_s \leq a'_{s+1} \leq s+1$. Thus, $a'_{s+1} = s$ or $a'_{s+1} = s+1$. If $a'_{s+1} = s+1$, then a new block begins at position $s+1$, contradicting the fact that $s+1$ lies inside π_j . Therefore, $a'_{s+1} = s$, so $P(2)$ holds.

Suppose now that for some r with $2 \leq r < \ell_j$, the statement $P(r)$ holds; that is,

$$(a'_s, a'_{s+1}, \dots, a'_{s+r-1}) = (s, s, s+1, \dots, s+r-2).$$

In particular, $a'_{s+r-1} = s+r-2$. We show that $P(r+1)$ holds. Since $\alpha^\uparrow$ is weakly increasing and α is a parking function,

$$s+r-2 = a'_{s+r-1} \leq a'_{s+r} \leq s+r.$$

Therefore, $a'_{s+r} \in \{s+r-2, s+r-1, s+r\}$. If $a'_{s+r} = s+r$, then a new block begins at position $s+r$. Since $r < \ell_j$, this position lies inside π_j , a contradiction. Thus, $a'_{s+r} \neq s+r$.

Now suppose that $a'_{s+r} = s+r-2$. By the induction hypothesis, all $r+1$ entries $a'_s, a'_{s+1}, \dots, a'_{s+r}$ lie between s and $s+r-2$. Therefore, the corresponding $r+1$ cars can park only among the r spots $s, s+1, \dots, s+r-1$ under the unit interval parking rule. By the pigeonhole principle, at least one car fails to park, contradicting $\alpha \in UPF_n$. Hence, $a'_{s+r} \neq s+r-2$.

The only remaining possibility is $a'_{s+r} = s+r-1$. Together with the induction hypothesis, this proves $P(r+1)$. Therefore, by induction, $P(r)$ holds for every $r = 2, 3, \dots, \ell_j$. Taking $r = \ell_j$, we conclude that

$$\pi_j = \begin{cases} (s), & \text{if } \ell_j = 1, \\ (s, s, s+1, s+2, \dots, s+\ell_j-2), & \text{if } \ell_j \geq 2. \end{cases}$$

□

Definition 1.6. *Weakly Increasing Parking Functions*

Let $\alpha = (a_1, a_2, \dots, a_n) \in [n]^n$. We say that α is weakly increasing if $a_i \leq a_{i+1}$ for all $i \in [n]$. We denote the weakly increasing rearrangement of any $\alpha =$

$(a_1, a_2, \dots, a_n) \in [n]^n$ by $\alpha^\uparrow = (a'_1, a'_2, \dots, a'_n)$. In particular, if α is a parking function, $\alpha^\uparrow$ satisfies $a_i \leq i$ for all $i \in [n]$.

Definition 1.7. *Inversion [3]*

An inversion of w is a pair of indices (i, j) such that $1 \leq i < j \leq n$ and $w_i > w_j$. We let $\text{Inv}(w)$ be the set of inversions of w and set $\text{inv}(w) = |\text{Inv}(w)|$.

Remark 1.1. *In our work, we find multiple connections to the Fibonacci numbers (add in the history to Pingala). For the interested reader we aggregate the list of results involving this famous sequence:*

- In Theorem A, we show the number of BLAH BLAHS are enumerated by the Fibonacci numbers.
- In Lemma B..... something

2. ENUMERATIVE RESULTS ON THE SETS

In section 2.1, we characterize and enumerate weakly increasing Cayley Permutations, Fubini Rankings, and Unit Interval Parking Functions, in that order. For each of these subsets of parking functions, we compare the cardinality of the weakly decreasing elements with the weakly increasing elements. In particular, we establish that

$$|\mathfrak{C}_n^\uparrow| = |\mathfrak{C}_n^\downarrow| = |\text{FR}_n^\uparrow| = |\text{FR}_n^\downarrow| = |\text{UPF}_n^\uparrow| \neq |\text{UPF}_n^\downarrow|.$$

In section 2.2, we characterize and enumerate the pairwise intersections of Cayley Permutations, Fubini Rankings, and Unit Interval Parking Functions. We establish that the weakly decreasing unit interval parking functions are enumerated by the Fibonacci numbers.

In section 2.3, we establish **Melissa: that which we will have established :-)** displacement of these three subsets of parking functions. **Melissa: Perhaps this section will be the grand-finale: bijection triangle that preserves inversions and possibly other nice things: acts as identity on intersections**

Melissa: Question: Do we use weakly increasing results for the bijections? If not, we could put the bijections first and then end with a fun Fibonacci surprise.

2.1. Weakly Increasing (and Decreasing) Cayley Permutations, Fubini Rankings, and Unit Interval Parking Functions. First, we focus on the weakly increasing and weakly decreasing subsets of Cayley permutations, Fubini rankings, and unit interval parking functions. We begin with a lemma regarding weakly increasing Cayley permutations.

Lemma 2.1. *For $n \geq 2$, an n -tuple $\alpha = (a_1, a_2, \dots, a_n) \in [n]^n$ is a weakly increasing Cayley permutation if and only if $a_1 = 1$ and $a_{i+1} - a_i$ is either 0 or 1 for each $i \in [n-1]$.*

Proof. ($\implies$) Let

$$\alpha = (a_1, a_2, \dots, a_n) \in \mathfrak{C}_n^\uparrow.$$

Since α is a weakly increasing Cayley permutation,

$$a_1 = 1 \quad \text{and} \quad a_1 \leq a_2 \leq \dots \leq a_n,$$

which implies that

$$a_{i+1} - a_i \geq 0 \quad \text{for all } 1 \leq i \leq n-1.$$

Suppose, for the sake of contradiction, that there exists an index i^* such that

$$a_{i^*+1} - a_{i^*} \geq 2.$$

Then

$$a_{i^*} < a_{i^*} + 1 < a_{i^*+1}.$$

Since α is a Cayley permutation and the value a_{i^*+1} appears in α , every positive integer less than a_{i^*+1} must also appear in α . In particular, $a_{i^*} + 1$ appears in α .

Because α is weakly increasing, any occurrence of $a_{i^*} + 1$ must appear after a_{i^*} and before a_{i^*+1} . Thus, there exists an index j satisfying

$$i^* < j < i^* + 1.$$

However, no such integer index exists, a contradiction.

Therefore,

$$a_{i+1} - a_i \leq 1 \quad \text{for all } 1 \leq i \leq n-1.$$

Combining this with

$$a_{i+1} - a_i \geq 0,$$

we conclude that

$$a_{i+1} - a_i \in \{0, 1\}$$

for all $1 \leq i \leq n-1$.

($\Leftarrow$) Suppose that

$$a_{i+1} - a_i \in \{0, 1\} \quad \text{for all } 1 \leq i \leq n-1.$$

Since each difference is nonnegative, we have

$$a_1 \leq a_2 \leq \cdots \leq a_n,$$

so α is weakly increasing.

We next show that α is a Cayley permutation. First recall that by assumption $a_1 = 1$.

Furthermore, since consecutive entries differ by at most 1, the sequence cannot skip any positive integer. Starting at $a_1 = 1$, the values that appear in α form the interval

$$\{1, 2, \dots, a_n\}.$$

Therefore, whenever an integer i appears in α , every integer $1 \leq j \leq i$ also appears in α . Hence,

$$\alpha \in \mathfrak{C}_n.$$

Since α is both weakly increasing and a Cayley permutation, we conclude that

$$\alpha \in \mathfrak{C}_n^\uparrow,$$

as desired. $\square$

Next, we prove the enumeration of each set's weakly increasing and weakly decreasing subsets. We begin by proving the enumeration for Cayley permutations, and similar logic applies for weakly Fubini rankings and weakly increasing unit interval parking functions. Weakly decreasing unit interval parking functions, on the other hand, are enumerated by the Fibonacci numbers.

Lemma 2.2. $|\mathfrak{C}_n^\uparrow| = |\mathfrak{C}_n^\downarrow| = 2^{n-1}$.

Proof. Let $\alpha^\uparrow = (a'_1, a'_2, \dots, a'_n)$ be the weakly increasing rearrangement of $\alpha \in \mathfrak{C}_n$.

We prove by induction on i that for each position i with $2 \leq i \leq n$, given any valid prefix $(a'_1, \dots, a'_{i-1})$, there are exactly 2 values a'_i can take that keep the sequence valid, namely $a'_i = a'_{i-1}$ or $a'_i = a'_{i-1} + 1$, and that the first position is always forced to be $a'_1 = 1$.

Suppose $i = 1$. By the Cayley condition, $\alpha^\uparrow$ must contain 1, and since $\alpha^\uparrow$ is weakly increasing, the first element must be the smallest possible value, which forces $a'_1 = 1$. There is only 1 option, so $\mathfrak{C}_1^\uparrow = \{(1)\}$ and $|\mathfrak{C}_1^\uparrow| = 2^{1-1} = 2^0 = 1$.

Now suppose $i = 2$. Since $a'_1 = 1$, the weakly increasing condition forces $a'_2 \geq 1$, but the Cayley rule prevents us from skipping directly to 3 because 2 has not appeared yet. Thus our choices are strictly limited to $a'_2 = 1$ or $a'_2 = 2$, giving exactly 2 options, so $\mathfrak{C}_2^\uparrow = \{(1, 1), (1, 2)\}$ and $|\mathfrak{C}_2^\uparrow| = 2^{2-1} = 2^1 = 2$.

Assume the claim holds for all positions up to k , so we have a valid prefix $(a'_1, a'_2, \dots, a'_k)$ where each choice follows the two conditions above. Now, we show that the next element a'_{k+1} has exactly 2 choices. The weakly increasing rule forces the lower bound $a'_{k+1} \geq a'_k$, since a'_{k+1} cannot be smaller than the previous term. Weakly increasing alone gives no upper bound, but the Cayley condition bounds it strictly at $a'_{k+1} \leq a'_k + 1$: if instead $a'_{k+1} = a'_k + 2$, the tuple would have a gap in value, violating the Cayley permutation condition.

For example, given the prefix $(1, 2, a'_3)$ with $a'_1 = 1, a'_2 = 2$, setting $a'_3 = a'_2 = 2$ gives $(1, 2, 2)$, a valid Cayley sequence; setting $a'_3 = a'_2 + 1 = 3$ gives $(1, 2, 3)$, also valid; but setting $a'_3 = a'_2 + 2 = 4$ gives $(1, 2, 4)$, which is not a Cayley sequence since 3 is missing.

Combining these two inequalities bounds a'_{k+1} strictly at $a'_k \leq a'_{k+1} \leq a'_k + 1$, and since $a'_{k+1} \in \mathbb{Z}$, this leaves exactly two possibilities: $a'_{k+1} = a'_k$, repeating the previous value, or $a'_{k+1} = a'_k + 1$, increasing the previous value by a maximum of 1.

By induction, every position $n \geq 2$ has exactly 2 independent choices, while the first position offers exactly 1. By the Fundamental Counting Principle,

$$|\mathfrak{C}_n^\uparrow| = 1 \cdot \underbrace{2 \cdot 2 \cdots 2}_{(n-1) \text{ times}} = 2^{n-1}.$$

Since Cayley permutations are invariant under permutations, we can reverse the tuples in $\mathfrak{C}_n^\uparrow$ to get $\mathfrak{C}_n^\downarrow$. So, $|\mathfrak{C}_n^\uparrow| = |\mathfrak{C}_n^\downarrow| = 2^{n-1}$. $\square$

Lemma 2.3. $|\text{FR}_n^\uparrow| = |\text{FR}_n^\downarrow| = 2^{n-1}$.

Finally, we enumerate the set of the weakly decreasing unit interval parking functions. We begin by establishing some facts that will help our proof.

Corollary 2.1. *For $\alpha \in \text{FR}_n \cup \text{UPF}_n$, if $a_i, \dots, a_k$ are repeats, they are not consecutive numbers.*

Corollary 2.2. *For a tuple of length n , up to $\lfloor n/2 \rfloor$ elements can repeat once.*

Proof. For an n -tuple with x elements that repeat once, there are $n - 2x$ places remaining. If $x > \lfloor n/2 \rfloor$, then $n - 2\lfloor n/2 \rfloor < 0$ places are left, which cannot happen. Thus, $x \leq \lfloor n/2 \rfloor$. $\square$

Let $\alpha = (a_1, a_2, \dots, a_n)$ be a tuple of length n . Let $\text{rev}(\alpha)$ be the operation that reverses the order of the elements in α such that $\text{rev}(\alpha) = (a_n, \dots, a_2, a_1)$. **Eban: does this go in a definition environment?**

Corollary 2.3. $\alpha^\uparrow = \text{rev}(\alpha^\downarrow)$ and $\alpha^\downarrow = \text{rev}(\alpha^\uparrow)$.

Corollary 2.4. For $a_i \in \alpha$, its corresponding $r_j \in \text{rev}(\alpha)$ is at index $j = n - i + 1$.

Lemma 2.4. If $\alpha \in \text{FR}_n \cap \text{UPF}_n$, then α is permutation invariant.

Proof. Since $\text{FR}_n \cap \text{UPF}_n := \text{UFR}_n$, we know cars have displacement of at most 1.

Moreover, by [4, Theorem 3.5], every block of $\alpha \in \text{UFR}_n$ is either of the form (i) or (i, i) . Recall that by [2, Theorem 2.9(2)], a rearrangement of α is in UPF_n if and only if the entries respect the relative order of the entries in each of the blocks. Since for $\alpha \in \text{UFR}_n$, the entries in each block are indistinguishable, it follows that any rearrangement of α respects the relative block order of the entries in each block. Thus, $\alpha \in \text{FR}_n \cap \text{UPF}_n$ is permutation invariant. $\square$

Lemma 2.5. $\alpha \in \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow$ if and only if $\text{rev}(\alpha) \in \text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow$.

Proof. Forward: Let $\alpha \in \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow$. Then, by Corollary 2.3, $\text{rev}(\alpha)$ is weakly increasing and $\text{rev}(\alpha) = \alpha^\uparrow$. By Lemma 2.4, $\text{rev}(\alpha) \in \text{FR}_n \cap \text{UPF}_n$ since Unit Fubini Rankings are permutation invariant. Thus $\text{rev}(\alpha) \in \text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow$.

Backwards: Let $\text{rev}(\alpha) \in \text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow$. Then by Corollary 2.3, α is weakly decreasing and $\text{rev}(\text{rev}(\alpha)) = \alpha^\downarrow$. By Lemma 2.4, $\alpha \in \text{FR}_n \cap \text{UPF}_n$ since Unit Fubini Rankings are permutation invariant. Thus $\alpha \in \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow$. $\square$

Lemma 2.6. $\text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow = \text{UPF}_n^\downarrow$.

Eban: needs illustration

Proof. Take $\alpha \in \text{UPF}_n^\downarrow$. We want to show that α is also a Fubini ranking such that $a_i = n - i$ or $a_i = n - i + 1$ for all $i \in [n]$, since α is weakly decreasing (see [1, Lemma 2.1] and Corollaries 2.3 and 2.4). For the sake of contradiction, suppose the Fubini condition is not satisfied. Let i^* be the smallest index at which the following occurs:

- Case 1: Say that $a_{i^*} < n - i^*$. If $i^* = n$ or $i^* = n - 1$, then $a_{i^*} < 1$, which implies $\alpha \notin \text{UPF}_n^\downarrow$, a contradiction. At car $i^* \in [n - 2]$, we have $n - i^*$ cars left to park, but the highest preference is less than $n - i^*$. Note that $\alpha \in \text{UPF}_n^\downarrow$ must satisfy $a_n = 1$. For the simplest case, suppose we want to ensure all remaining parking spots $s \in [n - i^* - 1]$ are occupied. Then, we have all $s \in [n - i^* - 1]$ listed in decreasing order. Since there are $n - i^*$ positions, at least one number, say j , repeats at least once. Note that we have at least two cars with preference j and at least one car with preference $j + 1$. Then, at least one car does not park because its preferred spot (j) and the spot right after ($j + 1$) are occupied. So, $\alpha \notin \text{UPF}_n^\downarrow$, a contradiction.
- Case 2: Say that $a_{i^*} > n - i^* + 1$. If $i^* = 1$, then $a_1 > n$, which implies $\alpha \notin \text{UPF}_n^\downarrow$, a contradiction. If $i^* = n$, then $a_n > 1$, so spot 1 is left empty, which implies $\alpha \notin \text{UPF}_n^\downarrow$, a contradiction. At car $1 < i^* < n$, we have $n - i^*$ cars left to park, but the highest preference among those cars is more than $n - i^* + 1$. To satisfy $a_n = 1$, we must skip at least one number. Then, there exists an $s \in [n - i^* - 1]$ such that spot s is left empty. So, $\alpha \notin \text{UPF}_n^\downarrow$, a contradiction.

So, $\alpha \in \text{UPF}_n^\downarrow$ satisfies the Fubini conditions for a weakly decreasing tuple. Then, $\alpha \in \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow$, which means $\text{UPF}_n^\downarrow \subseteq \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow$. Also note $\text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow \subseteq \text{UPF}_n^\downarrow$. Thus, $\text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow = \text{UPF}_n^\downarrow$. $\square$

Consider these examples:

- Case 1: Consider $(7, 5, 5, 2, 2, 1, 1)$. At $i^* = 4$, we have $a_4 = 2 < 7 - 4$. By the weakly decreasing condition, we have conflicting preferences (and also spot $i^* = 4$ is left empty).
- Case 2: Consider $(7, 6, 6, 5, 4, 3, 1)$. We have $i^* = 3$ and $a_3 = 6 > 7 - 3 + 1$. To reach 1, we had to skip spot 2 (and also there's a conflict with the second $6 > 7 - 3 + 1$ at $i^* = 3$).

The following result was proven for $\text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow$ in [4, Theorem 3.8]. Here, we extend those findings to the weakly decreasing set and intersection and present an independent proof.

Lemma 2.7. $|\text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow| = |\text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow| = |\text{UPF}_n^\downarrow| = F_{n+1}$.

Proof. For a tuple in one of these three sets ($\text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow$, $\text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow$, or $\text{UPF}_n^\downarrow$), by Corollary 2.2, we can choose $0 \leq k \leq \lfloor n/2 \rfloor$ elements, a_i , to be repeats. By Corollary 2.1, the total number of elements from which we can choose to be repeats is $n - k$ (subtracting the k numbers that are greater than and consecutive to each a_i). Adding, we have

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k}$$

tuples in each set. This summation yields the Fibonacci sequence (with offset of 1) [5, A000045] with terms for $1 \leq n \leq 7$ being

$$1, 2, 3, 5, 8, 13, 21.$$

□

2.2. Intersections of Fubini Rankings, Unit Interval Parking Functions, and Cayley Permutations. In this section, we characterize the pairwise and triple intersections of $\mathfrak{C}_n$, FR_n , and UPF_n . As in section 2.1, the weakly increasing rearrangement $\alpha^\uparrow$ is the main tool here, with each intersection corresponding to a specific structural condition on $\alpha^\uparrow$. We also give exact counts for each intersection. We begin with the intersection of Cayley permutations and Fubini rankings

Lemma 2.8 (Characterization of Cayley Fubini Rankings). *Let $n \geq 1$. Then $\alpha \in [n]^n$ is in $\mathfrak{C}_n \cap \text{FR}_n$ if and only if the weakly increasing rearrangement of α , denoted by $\alpha^\uparrow = (a'_1, a'_2, \dots, a'_n)$, satisfies $a'_i = i$ for all $1 \leq i < \max(\alpha)$ and $a'_j = \max(\alpha)$ for all $\max(\alpha) \leq j \leq n$.*

Corollary 2.5. $|\mathfrak{C}_n^\uparrow \cap \text{FR}_n^\uparrow| = |\mathfrak{C}_n^\downarrow \cap \text{FR}_n^\downarrow| = n$.

Theorem 2.1. $|\mathfrak{C}_n \cap \text{FR}_n| = n! \sum_{k=1}^n \frac{1}{k!}$.

Proof. By Lemma 2.8 and Corollary 2.5, tuples $\alpha \in \mathfrak{C}_n \cap FR_n$ have the following forms:

$$\text{all permutations of } \left\{ \begin{array}{l} \underbrace{(1, 1, \dots, 1)}_{n\text{-times}} \\ (1, \underbrace{2, \dots, 2}_{(n-1)\text{-times}}) \\ \vdots \\ (1, 2, \dots, \underbrace{a'_i, \dots, a'_i}_{(n-a'_i+1)\text{-times}}) \\ \vdots \\ (1, 2, \dots, n). \end{array} \right.$$

Note that there are n distinct forms, and all permutations are valid since Cayley permutations and Fubini rankings are permutation invariant. For an $(n+1)$ -tuple $\gamma \in \mathfrak{C}_{n+1} \cap FR_{n+1}$ has one new distinct form (i.e., a new tuple in which $n = \gamma_i$ can be repeated).

$$\begin{aligned} (1) &\in \mathfrak{C}_1 \cap FR_1 \\ (1, 1), (1, 2) &\in \mathfrak{C}_2 \cap FR_2 \\ (1, 1, 1), (1, 2, 2), (1, 2, 3) &\in \mathfrak{C}_3 \cap FR_3 \\ (1, 1, 1, 1), (1, 2, 2, 2), (1, 2, 3, 3), (1, 2, 3, 4) &\in \mathfrak{C}_4 \cap FR_4 \\ &\vdots \\ \underbrace{(1, \dots, 1)}_{n\text{-times}}, \dots, (1, 2, \dots, \underbrace{a_i, \dots, a_i}_{(n-a_i+1)\text{-times}}), (1, 2, 3, \dots, n) &\in \mathfrak{C}_n \cap FR_n \end{aligned}$$

For a tuple in which all elements are distinct, there are $n!$ permutations. If there are two distinct elements, there are $\binom{n}{2}(n-2)!$ permutations. If there are k distinct elements, there are $\binom{n}{k}(n-k)!$ permutations. Adding, we have

$$\begin{aligned} |\mathfrak{C}_n \cap FR_n| &= n! + \binom{n}{2}(n-2)! + \dots + \binom{n}{1}(n-(n-1))! + \binom{n}{n}(n-n)! \\ &= n! + \frac{n!}{2!} \cdot \frac{(n-2)!}{(n-2)!} + \dots + \frac{n!}{1!(n-1)!} + 1 \\ &= n! + \frac{n!}{2!} + \dots + \frac{n!}{(n-1)!} + 1 \\ &= n! \left(\frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{(n-1)!} + \frac{1}{n!} \right) \\ &= n! \sum_{k=1}^n \frac{1}{k!} \end{aligned}$$

as claimed. □

Computationally, the sequence for $\mathfrak{C}_n \cap FR_n$ is [5, A002627], whose terms for $1 \leq n \leq 7$ are:

$$1, 3, 10, 41, 206, 1237, 8660.$$

Second, we characterize the intersection of Cayley permutations and unit interval parking functions. Then, we establish the enumeration of the weakly increasing, weakly decreasing, and general intersections.

Lemma 2.9 (Characterization of Cayley Unit Interval Parking Functions). *Let $n \geq 1$. Then $\alpha \in [n]^n$ lies in $\mathfrak{C}_n \cap \text{UPF}_n$ if and only if its weakly increasing rearrangement $\alpha^\uparrow = (a'_1, a'_2, \dots, a'_n)$ satisfies the following:*

- (1) $a'_1 = 1$ and $a'_{k+1} - a'_k \leq 1$ for all $1 \leq k < n$ (the Cayley condition).
- (2) At most one value among the a'_i repeats.
- (3) If $a'_i = a'_{i+1}$, then $a'_{i+2} = a'_i + 1$ (Right after the repeat, step up by one exactly).
- (4) If $a'_i = a'_{i+1}$, then in the original α every entry $a_j > a'_i$ must occur after both occurrences of a'_i in α , and those entries occur in strictly increasing order.

Proof. For reference, we recall the following definition from the paper:

[2, Definition 2.7] Given $\alpha = (a_1, \dots, a_n) \in \text{PF}_n$, let $\alpha^\uparrow = (a'_1, \dots, a'_n)$ be its weakly increasing rearrangement. The partition of $\alpha^\uparrow$ into a concatenation $\alpha' = \pi_1 | \pi_2 | \dots | \pi_m$, where each block π_j begins at (and includes) the j -th entry satisfying $a'_i = i$, is called the *block structure* of α . Each π_j is called a *block* of α . A new block starts at every position i where the sorted value satisfies $a'_i = i$.

For example, if $\alpha = (8, 1, 5, 5, 1, 2, 4, 7) \in \text{PF}_8$, then $\alpha^\uparrow = (1, 1, 2, 4, 5, 5, 7, 8)$ and its block structure is $\alpha' = 112|4|55|7|8$.

We first prove the forward direction. Assume $\alpha \in \mathfrak{C}_n \cap \text{UPF}_n$. Let $\alpha^\uparrow = (a'_1, \dots, a'_n)$ be its weakly increasing rearrangement.

For Condition (1), we must show $a'_1 = 1$ and $a'_{k+1} - a'_k \leq 1$ for all $1 \leq k < n$, the Cayley condition. By definition of $\mathfrak{C}_n$, this condition holds.

For Condition (2), we must show at most one value among the a'_i repeats. Suppose, for contradiction, that at least two distinct values repeat in $\alpha^\uparrow$. By Condition (1), $\alpha^\uparrow$ is non-decreasing with step size at most 1. Let i^* be the smallest index at which a repeat occurs, which is the first place where $a'_i = a'_{i+1}$. Since $\alpha \in \text{UPF}_n$, the entries from this point onward, together with the repeated value, must themselves form a single final block that is a valid element of UPF_n on its own. By Lemma 1.1, the only valid shape for a single final block is the canonical form

$$[i^*, i^*, i^* + 1, \dots, n - 1]$$

which is one repeated value i^* followed by a strictly increasing run $i^* + 1, i^* + 2, \dots$ with step size exactly 1 and no further repeats. This comes from the displacement at most 1 rule: once a value repeats, every subsequent slot is fully determined, leaving no room for a second repeat anywhere later in the block. But our assumption requires a second distinct repeated value to occur somewhere in this same block and no such shape exists. This contradicts the base case of Theorem 2.9, thus our assumption that two distinct values repeat must be false. Therefore, at most one value repeats in $\alpha^\uparrow$.

For Condition (3), we must show that if $a'_i = a'_{i+1}$, then $a'_{i+2} = a'_i + 1$. By Condition (2) there is at most one repeated value i^* , appearing exactly twice. This

pair lies at the start of the final block. By the single-block form in the base case of [2, Theorem 2.9], the element immediately after the pair must be exactly one greater than i^* . Therefore $a'_{i+2} = a'_i + 1$.

For Condition (4), we must show that if $a'_i = a'_{i+1}$, then in the original α every entry $a_j > a'_i$ occurs after both occurrences of a'_i , and those entries occur in strictly increasing order. Since the repeated value $i^* = a'_i$ and all larger entries belong to the same final block of $\alpha^\uparrow$. By [2, Theorem 2.9 Part (2)], since $\alpha \in \text{UPF}_n$, the sequence α must respect the relative order of entries inside each block. Inside the canonical final block the order is $i^*, i^*, i^* + 1, i^* + 2, \dots$. Therefore both copies of i^* must precede every larger entry in α , and the larger entries must appear in strictly.

We now prove the backward direction. Assume $\alpha \in [n]^n$ whose weakly increasing rearrangement $\alpha^\uparrow$ satisfies Conditions (1)–(4).

By Condition (1), $\alpha \in \mathfrak{C}_n$. It remains to show $\alpha \in \text{UPF}_n$. By Conditions (1)–(3), the block structure of $\alpha^\uparrow$ consists of the initial singleton blocks

$$1 \mid 2 \mid \dots \mid (i^* - 1)$$

(if a repeat i^* exists) followed by a single final block whose sorted form is exactly the Lemma 1.1 canonical single-block unit interval parking function

$$[i^*, i^*, i^* + 1, \dots, n - 1]$$

(or simply $[i^*, i^*]$ if the repeat is at the end). Each singleton block is trivially in UPF_n for its size. The final block is a shifted copy of the unique canonical single-block element of UPF_n (base case of [2, Theorem 2.9]), thus valid on its own consecutive spots.

Condition (4) guarantees that α respects the relative order of entries inside this final block.

By Theorem 2.9 Part (1) [2], smaller valid pieces of α can be merged together. Specifically, we can interleave the blocks while preserving the order of entries inside each individual block, and the combined result is still a unit interval parking function. Thus $\alpha \in \text{UPF}_n$.

Both directions hold, so $\alpha \in \mathfrak{C}_n \cap \text{UPF}_n$ if and only if Conditions (1)–(4) are satisfied on $\alpha^\uparrow$. $\square$

Corollary 2.6. *Let $\alpha \in \mathfrak{C}_n \cap \text{UPF}_n$. Then $\alpha^\uparrow$ is strictly increasing except possibly for one repeated value i^* appearing exactly twice consecutively, immediately followed by $i^* + 1$. In the original sequence α , both occurrences of i^* comes before every larger entry, and the larger entries appear in strictly increasing order.*

Corollary 2.7. $|\mathfrak{C}_n^\uparrow \cap \text{UPF}_n^\uparrow| = n$.

Corollary 2.8. For $n = 1$, $|\mathfrak{C}_n^\downarrow \cap \text{UPF}_n^\downarrow| = 1$. For $n \geq 2$, $|\mathfrak{C}_n^\downarrow \cap \text{UPF}_n^\downarrow| = 2$.

Eban: note that this is the same for the triple intersection (both weakly increasing and weakly decreasing). however with the triple intersection, it is easier to prove this because of our lemma. So should this result be moved down there? It might break the consistency of having each intersection together though

Interestingly, our next theorem states that $\mathfrak{C}_n \cap \text{UPF}_n$ is enumerated by the same sequence [5, A002627] as $\mathfrak{C}_n \cap \text{FR}_n$.

Theorem 2.2. $|\mathfrak{C}_n \cap \text{UPF}_n| = n! \sum_{k=1}^n \frac{1}{k!}$.

Proof. By Lemma 2.9 and Corollary 2.7, tuples $\alpha \in \mathfrak{C}_n \cap UPF_n$, have the following forms:

$$\text{certain permutations of } \left\{ \begin{array}{l} (1, 1, 2, 3, \dots, n-2, n-1) \\ (1, 2, 2, 3, \dots, n-2, n-1) \\ \vdots \\ (1, 2, 3, \dots, a_i, a_i, \dots, n-2, n-1) \\ \vdots \\ (1, 2, 3, \dots, n-2, n-2, n-1) \\ (1, 2, 3, \dots, n-2, n-1, n-1) \\ (1, 2, 3, \dots, n-2, n-1, n). \end{array} \right.$$

To calculate the size of this intersection, we calculate the total number of permutations for each form and remove (by dividing out) the number of invalid ones. **Eban: this logic is right but not sure it matches up with table in the best way** Table 1 shows the number of valid permutations based on the element a_i repeated.

a_i repeated	#valid perms
1	1
2	n
3	$n(n-1)$
$\vdots$	$\vdots$
a_i	$n!/(n-a_i+1)!$
$\vdots$	$\vdots$
$n-2$	$n!/3!$
$n-1$	$n!/2!$
none	$n!$

TABLE 1. Number of permutations based on the repeated element

Adding, we have

$$\begin{aligned} |\mathfrak{C}_n \cap UPF_n| &= n! \sum_{k=1}^n \frac{1}{k!} \\ &= n! \left(\frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{(n-2)!} + \frac{1}{(n-1)!} + \frac{1}{n!} \right) \\ &= n! + \frac{n!}{2!} + \dots + n(n-1) + n + 1 \end{aligned}$$

as claimed. $\square$

Next, we characterize the intersection of Fubini rankings and unit interval parking functions. Then, we establish the enumeration of the weakly increasing, weakly decreasing, and general intersections.

Lemma 2.10 (Characterization of Fubini Unit Interval Parking Functions). *Let $n \geq 1$. Then, $\alpha \in [n]^n$ is in $\text{FR}_n \cap \text{UPF}_n$ if and only if the weakly increasing rearrangement of α , denoted by $\alpha^\uparrow = (a'_1, a'_2, \dots, a'_n)$, satisfies the following:*

- (1) either $a'_i = i$ or $a'_i = a'_{i-1}$ for all $1 \leq i \leq n$, and
(2) if a'_i repeats, it repeats once.

Now, we establish the enumeration of the general intersection of Fubini rankings and unit interval parking functions. The enumeration was proven in [1] by bijection to be the sequence [5, A080599], whose terms for $1 \leq n \leq 7$ are:

$$1, 3, 12, 66, 450, 3690, 35280.$$

We present a closed formula with an independent proof here.

Theorem 2.3. $|\text{FR}_n \cap \text{UPF}_n| = n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{2^k \cdot k!} \frac{(n-k)!}{(n-2k)!}.$

Proof. By Lemma 2.10, tuples $\alpha \in \text{FR}_n \cap \text{UPF}_n$ have the following forms:

$$\text{all permutations and certain variations of } \begin{cases} (1, 2, \dots, n-2, n-1, n) \\ (1, \dots, a_i, a_i, a_i+2, \dots, n-2, n-1, n) \\ (1, \dots, a_i, a_i, \dots, a_j, a_j, \dots, n-1, n) \\ (1, \dots, a_i, a_i, \dots, a_j, a_j, \dots, n-2, n-1, n-1) \\ (1, \dots, a_i, a_i, \dots, a_j, a_j, \dots, a_k, a_k, \dots, n-1, n) \\ (1, \dots, a_i, a_i, \dots, a_j, a_j, \dots, a_k, a_k, \dots, n-2, n-1, n-1) \\ \vdots \\ \text{up to } \lfloor n/2 \rfloor \text{ } a_i \in \alpha \text{ repeat once by Corollary 2.2.} \end{cases}$$

Eban: cases does not fit within margin...

The number of variations is limited by the fact that only nonconsecutive a_i 's can repeat (see Corollary 2.1). We can calculate the number of permutations and variations to get the count of our intersection. These values depend on the number of repeated elements, as shown in Table 2. **Eban: does the table math need more explanation?**

$\#a_i \text{ rep}$	$\#\text{perms} \cdot \#\text{vars}$
0	$n! \cdot 1$
1	$\binom{n}{2}(n-2)! \cdot (n-1)$
2	$\binom{n}{2}\binom{n-2}{2}(n-4)! \cdot \binom{n-2}{2}$
3	$\binom{n}{2}\binom{n-2}{2}\binom{n-4}{2}(n-6)! \cdot \binom{n-3}{3}$
$\vdots$	$\vdots$
k	$\binom{n}{2}\binom{n-2}{2} \cdots \binom{n-2(k-1)}{2}(n-2k)! \cdot \binom{n-k}{k}$
$\vdots$	$\vdots$
$\lfloor n/2 \rfloor$	$\binom{n}{2}\binom{n-2}{2} \cdots \binom{n-2(\lfloor n/2 \rfloor - 1)}{2}(n-2\lfloor n/2 \rfloor)! \cdot \binom{n-\lfloor n/2 \rfloor}{\lfloor n/2 \rfloor}$

TABLE 2. Number of permutations and variations based on the number of repeats

Adding the values from the table above, we have

$$\begin{aligned}
& n! + \binom{n}{2}(n-2)!\binom{n-1}{1} + \binom{n}{2}\binom{n-2}{2}(n-4)!\binom{n-2}{2} \\
& \quad + \binom{n}{2}\binom{n-2}{2}\binom{n-4}{2}(n-6)!\binom{n-3}{3} + \cdots + \text{last term} \\
&= n! + \frac{n!}{2!(n-2)!}(n-2)!(n-1) + \frac{n!}{2!(n-2)!} \frac{(n-2)!}{2!(n-2-2)!}(n-4)! \frac{(n-2)!}{2!(n-2-2)!} \\
& \quad + \frac{n!}{2!(n-2)!} \frac{(n-2)!}{2!(n-2-2)!} \frac{(n-4)!}{2!(n-4-2)!}(n-6)! \frac{(n-3)!}{3!(n-3-3)!} + \cdots + \text{last term} \\
&= n! + \frac{n!(n-1)}{2} + \frac{n!(n-2)!}{8(n-4)!} + \frac{n!(n-3)!}{48(n-6)!} + \cdots + \text{last term} \\
&= n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{2^k \cdot k!} \frac{(n-k)!}{(n-2k)!}
\end{aligned}$$

as claimed. □

Finally, we characterize the intersection of Cayley permutations, Fubini rankings, and unit interval parking functions. Then, we establish the enumeration of the weakly increasing, weakly decreasing, and general intersections.

Lemma 2.11 (Characterization of Cayley Fubini Unit Interval Parking Functions). *Let $n \geq 1$. Then $\alpha \in [n]^n$ is in $\mathfrak{C}_n \cap \text{FR}_n \cap \text{UPF}_n$ if and only if the weakly increasing rearrangement of α , denoted by $\alpha^\uparrow = (a'_1, a'_2, \dots, a'_n)$, satisfies $a'_i = i$ for all $1 \leq i \leq \max(\alpha)$, $a'_j = \max(\alpha)$ for all $\max(\alpha) \leq j \leq n$, and $n-1 \leq \max(\alpha) \leq n$.*

Corollary 2.9. $\mathfrak{C}_n^\downarrow \cap \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow = \mathfrak{C}_n^\downarrow \cap \text{UPF}_n^\downarrow$.

Corollary 2.10. For $n = 1$, $|\mathfrak{C}_n^\uparrow \cap \text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow| = |\mathfrak{C}_n^\downarrow \cap \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow| = 1$. For $n \geq 2$, $|\mathfrak{C}_n^\uparrow \cap \text{FR}_n^\uparrow \cap \text{UPF}_n^\uparrow| = |\mathfrak{C}_n^\downarrow \cap \text{FR}_n^\downarrow \cap \text{UPF}_n^\downarrow| = 2$.

Theorem 2.4. $|\mathfrak{C}_n \cap \text{FR}_n \cap \text{UPF}_n| = \left\lfloor \frac{3n!}{2} \right\rfloor$.

Proof. By Lemma 2.11 and Corollaries 2.9 and 2.10, tuples $\alpha \in \mathfrak{C}_n \cap \text{FR}_n \cap \text{UPF}_n$, have the following forms:

$$\text{all permutations of } \begin{cases} (1, 2, \dots, n-2, n-1, n) \\ (1, 2, \dots, n-2, n-1, n-1). \end{cases}$$

Calculating the number of permutations, we have

$$|\mathfrak{C}_n \cap \text{FR}_n \cap \text{UPF}_n| = n! + \binom{n}{2}(n-2)! = n! + \frac{n!}{2(n-2)!}(n-2)! = \frac{3n!}{2}.$$

The floor function is necessary for $n = 1$ since $(1-2)!$ is undefined. Thus,

$$|\mathfrak{C}_n \cap \text{FR}_n \cap \text{UPF}_n| = \left\lfloor \frac{3n!}{2} \right\rfloor$$

as claimed. □

The sequence for $|\mathfrak{C}_n \cap FR_n \cap UPF_n|$ is given by [5, [A070960](#)]. The first seven terms are:

$$1, 3, 9, 36, 180, 1080, 7560.$$

Having established the pairwise and triple intersections of $\mathfrak{C}_n$, FR_n , and UPF_n , we now turn to a different statistic on these sequences: displacement, which measures how far each entry sits from its assigned or “ideal” position.

3. BIJECTIONS AND INVERSIONS

This section should contain the proofs that the sets are in bijection and that the bijections preserve the number of inversions.

4. ON THE DISPLACEMENT STATISTIC

5. FUTURE WORK.

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